



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade Awarded Category - I by UGC

PRN 2507052

Institute : (0705) SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme : (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch : 2025-29

Semester : I

Course Name : Programming in C

Course Code : 0705210101

Date : 27/09/2025

Maximum Marks : 15

Day : Saturday

Time : 11:30 AM to 12:30 PM

NOTE: DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

1. Attempt all questions. Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
2. Write both the correct option number and the answer text. (for MCQ question)
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

Q. No.		Marks	CO
Q1	Apply the knowledge of C language to check error segmentation fault also occurs: (Choose one). Give justification. 1. Because scanf requires the memory address of the variable, and without & it tries to write to an invalid location. 2. Because scanf can only work with integers, not other data types. 3. Because scanf automatically allocates memory if & is missing. 4. Because scanf requires pointers only when working with arrays.	1	CO1
Q2	Define the use of # in programming in c with justification 1. It is used for memory allocation at runtime. 2. It is used to define preprocessor directives before compilation. 3. It is used to terminate a statement. 4. It is used to represent comments in the program.	1	CO1
Q3	Analyze the output, what will be value of c? (0.5 mark). Give Dry run (0.5 mark). int main() { int a=100,b=200,c; c=(a==100 b>200); printf("c=%d",c); } 1. c = 0 2. c = 1 3. c = 100 4. c = 200	1	CO1
Q4	What is the output? (0.5 mark), Give dry run (0.5 mark). #include <stdio.h> int main () {	1	CO1



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PRN 25070521228

Institute : (0705) SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme : (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch : 2025-29

Semester : I

Course Name : Programming in C

Course Code : 0705210101

Date : 27/09/2025

Maximum Marks : 15

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NOTE: DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

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Q. No.		Marks	CO	BL
Q1	Apply the knowledge of C language to check error segmentation fault also occurs: (Choose one). Give justification. 1. Because scanf requires the memory address of the variable, and without & it tries to write to an invalid location. 2. Because scanf can only work with integers, not other data types. 3. Because scanf automatically allocates memory if & is missing. 4. Because scanf requires pointers only when working with arrays.	1	CO1	BL3
Q2	Define the use of # in programming in c with justification 1. It is used for memory allocation at runtime. 2. It is used to define preprocessor directives before compilation. 3. It is used to terminate a statement. 4. It is used to represent comments in the program.	1	CO1	BL1
Q3	Analyze the output, what will be value of c? (0.5 mark). Give Dry run (0.5 mark). int main() { int a=100,b=200,c; c=(a==100 b>200); print("c=%d",c); } 1. c = 0 2. c = 1 3. c = 100 4. c = 200	1	CO1	BL4
Q4	What is the output? (0.5 mark), Give dry run (0.5 mark). #include <stdio.h> int main () {	1	CO1	BL1

```
int x = 5;
int y = ++x * 2;
printf("y = %d", y);
return 0;
}
```

1. y = 10
2. y = 12
3. y = 11
4. y = 5

1

CO1

BL1

Q5 Match the following C programming symbols with their correct meaning:

Symbol	Meaning Options
a) #	1. Statement terminator
b) <	2. Integer keyword
c) ;	3. Used in header file inclusion
d) //	4. Preprocessor directive symbol
e) int	5. Single-line comment

- ☒ 1. a-4, b-3, c-1, d-5, e-2
- ☐ 2. a-3, b-4, c-2, d-1, e-5
- ☐ 3. a-1, b-2, c-3, d-4, e-5
- ☐ 4. a-2, b-1, c-4, d-3, e-5

Q6 What will be the output of this program? (0.5 mark) Give dry run. (0.5 mark).

1

CO2

BL2

```
#include <stdio.h>
int main() {
    int arr[5] = {1, 2, 3, 4, 5};
    printf("%d", arr[2]);
    return 0;
}
```

1. 1
- ☒ 2. 2
- ☐ 3. 3
4. 5

Q7 Which function is used to find the length of a string in C? (0.5 mark) Write one example program (0.5 mark).

1

CO2

BL1

- ☒ 1. strlen()
2. strcpy()
3. strcmp()
4. strcat()

use the code and give the output (0.5 mark) Give dry run. (0.5 mark).

1

CO2

BL4

```
#include <stdio.h>
#include <string.h>

int main() {
    char str1[] = "Hello";
    char str2[] = "World";
    printf("%s", strcat(str1, str2));
    return 0;
}
```

- 1. Hello
- 2. World
- 3. HelloWorld
- 4. Error

Q9 What will strcmp("Apple", "Apple") return? (0.5 mark)

1

CO2

BL1

Give dry run. (0.5 mark).

- 1. 1
- 2. -1
- 3. 0
- 4. Error

Q10 Apply the knowledge and complete the missing line in the program to read a string using fgets ().

1

CO2

BL3

```
#include <stdio.h>
int main() {
    char str[50];
    printf("Enter a string: ");
```

Missing line here

```
printf ("You entered: %s", str);
return 0;
}
```

Q11 Write the difference between While & Do While with a code example.

2.5

CO1

BL4

Q12 Write a C program, to declare array of size 5, take input values from user and print the sum of all the values of the array.

2.5

CO2

BL4



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Seat No.

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Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch: 2022-26,2023-27,2024-28,2025-29

Semester: I

Course: Programming in C

Course Code: 0705210107

Date: 10/12/2025

Maximum Marks: 45

Day: Wednesday

Time: 10:00 - 11:30

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

- Attempt all questions.
- Figures to the right indicate maximum marks.
- Non programmable calculator is allowed.
- Make suitable assumptions wherever necessary.

- Q.1
- Differentiate between While, Do-While, and For loops in C based on condition checking, execution flow, and usage. 5 CO1 BL2
 - 1.b.1. Write a C program to demonstrate the use of logical operators (&&, ||, !) with a real-life example of checking student scholarship eligibility. In a college, scholarship eligibility is based on marks and attendance as per the following conditions: 2.5 CO1 BL2
 - The student should have marks greater than 80.
 - The student should have attendance greater than 75%.
 - If both conditions are true, the student is eligible for a scholarship.
 - If the student has marks less than 40 or attendance less than 50%, the student is at risk of failing.
 - If the student's marks are not less than 80, display that the student has scored well above average.

The program should check all conditions using logical AND (&&), OR (||), and NOT (!) operators and display appropriate messages.

- 1.b.2. Write a C program to demonstrate the working of unary operators (++ and --) using a real-life example. 2.5 CO1 BL2

Assume there are a certain number of visitors in a park.

- When a visitor enters the park, the total number of visitors increases by one (use the increment operator).
- When a visitor leaves the park, the total number of visitors decreases by one (use the decrement operator).

The program should display the number of visitors before and after each operation, showing the difference between pre-increment and post-decrement.

- | | | | |
|---|--|---|---------|
| 2 | <p>a) Apply concept of loops and write a C program to print the following pattern using nested loops:</p> <pre>***** **** *** ** *</pre> | 5 | CO2 BL3 |
| | <p>b) Identify the use of five commonly used string functions from the <string.h> library and write the program.</p> <p>The program should display the result after performing each operation.</p> | 5 | CO2 BL3 |
| 3 | <p>a) Identify the use and explain the following terms in C programming with suitable examples:</p> <ul style="list-style-type: none"> • Actual Parameter • Formal Parameter • Local Variable • Global Variable • User-defined and Predefined Functions | 5 | CO3 BL3 |
| | <p>b) Build two C programs to swap the values of two variables:</p> <ul style="list-style-type: none"> • Using Call by Value (2marks) • Using Call by Reference (2marks) <p>In the first program (Call by Value), the original values will not change because only copies are passed.</p> <p>In the second program (Call by Reference), the original values will change because addresses are passed. (1 mark)</p> | 5 | CO3 BL3 |
| 4 | <p>a) Analyze and write a structure in C to store student records containing:</p> <ul style="list-style-type: none"> • Name • Roll number • Marks of 3 subjects | 5 | CO4 BL4 |

b) Analyze and explain the concept of structure write a C program using a nested structure to store and display student information such as name, roll number, and address (containing city and pin code). 5 CO4 BL4

5 a) Write a function to compute the area and circumference of a circle, having area and circumference as pointer arguments and radius as an ordinary argument. 5 CO5 BL5

Input Format:

A single line of input contains a float value representing the radius of the circle.

Output Format:

First line: Displays the area of the circle as a floating-point value formatted to 2 decimal places.

Second Line: Displays the perimeter (circumference) of the circle as a floating-point value formatted to 2 decimal places.

Note: Take $\pi = 3.14$

OR

b) Compare between `malloc()` and `calloc()` functions in C programming. 5 CO5 BL5



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PRN 25070521228

Exam : MID SEMESTER EXAMINATION (CA-II)

Institute : (0705) SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme : (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch : 2025-29

Semester : I

Course Name : Mathematics-I

Course Code : 0705210103

Date : 29/09/2025

Day : Monday

Maximum Marks : 20

Time : 11:30 AM to 12:30 PM

NOTE: DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

1. Attempt **FOUR** questions. Use Ball Pen/ Ink Pen (Blue only) to answer all questions
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Solve any **FOUR** of the following questions.

Q. No.		Marks	CO	BL
Q1	Find the rank of the matrix A by reducing it in echelon form $A = \begin{bmatrix} 3 & 4 & -1 & -6 \\ 2 & 3 & 2 & -3 \\ 2 & 1 & -14 & -9 \\ 1 & 3 & 13 & 3 \end{bmatrix}$	5	CO4	BL2
Q2	Test the consistency of following system of linear equations and solve if consistent $5x + 3y + 7z = 4, 3x + 26y + 2z = 9, 7x + 2y + 10z = 5$	5	CO4	BL3
Q3	Show that the equations $3x + 4y + 5z = a, 4x + 5y + 6z = b, 5x + 6y + 7z = c$, do not have a solution unless a, b, c are in Arithmetic Progression.	5	CO4	BL3
Q4	Verify Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence compute A^{-1}	5	CO5	BL4
Q5	Find the eigen values, eigen vectors for the given matrix $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$ Also, find dimensions.	5	CO5	BL4



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Seat No. 625006

Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Programme: (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch: 2023-27, 2024-28, 2025-29

Semester: I

Course: Mathematics-I

Course Code: 0705210102

Date: 05/12/2025

Day: Friday

Maximum Marks: 45

Time: 10:00 - 11:30

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

- Attempt all questions.
- Figures to the right indicate maximum marks.
- Non programmable calculator is allowed.
- Make suitable assumptions wherever necessary.

Q.1 Solve any TWO

- Apply Leibnitz's Rule to prove that
 $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 + a^2)y_n = 0$ if $y = e^{a \sin^{-1} x}$.
6 CO1 BL3
- Apply Taylor's theorem to expand $\log(\cos x)$ about point $\frac{\pi}{3}$.
6 CO2 BL3
- Solve i) $\lim_{x \rightarrow \pi/2} (\sin x)^{\tan x}$ ii) $\lim_{x \rightarrow 0} \frac{xe^x - \log(1+x)}{x^2}$
6 CO2 BL3

Q.2 Solve any TWO

- If $x + y + z = u$, $y + z = uv$, $z = uvw$, prove that $\frac{\partial(x,y,z)}{\partial(u,v,w)} = u^2v$.
6.5 CO3 BL5
- Evaluate the point on the plane $ax + by + cz = p$ at which the function $f = x^2 + y^2 + z^2$ has minimum value and also estimate the value of the function f at this point.
6.5 CO3 BL5
- If $u = \log\left(\frac{x^3 + y^3}{x^2 + y^2}\right)$, Prove that
i) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 1$ ii) $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = -1$
6.5 CO3 BL5